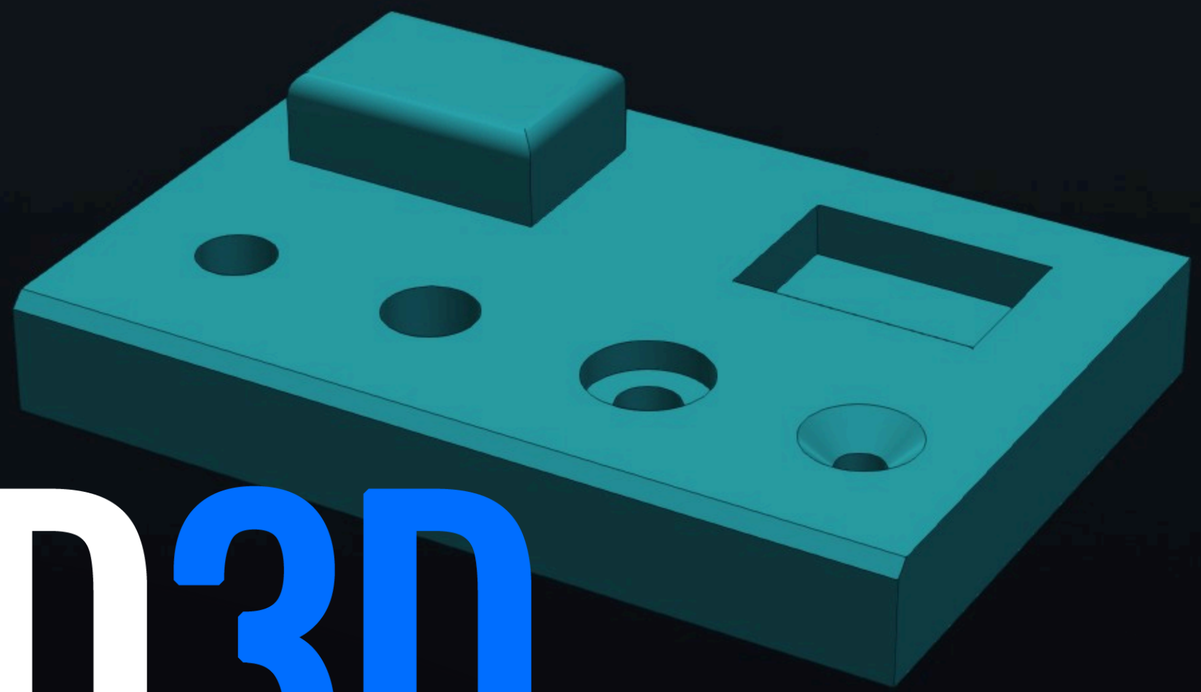
INVESTOR & PARTNER BRIEFING · OCTOBER 2026



VACAD3D

Real scans in. Editable CAD out.

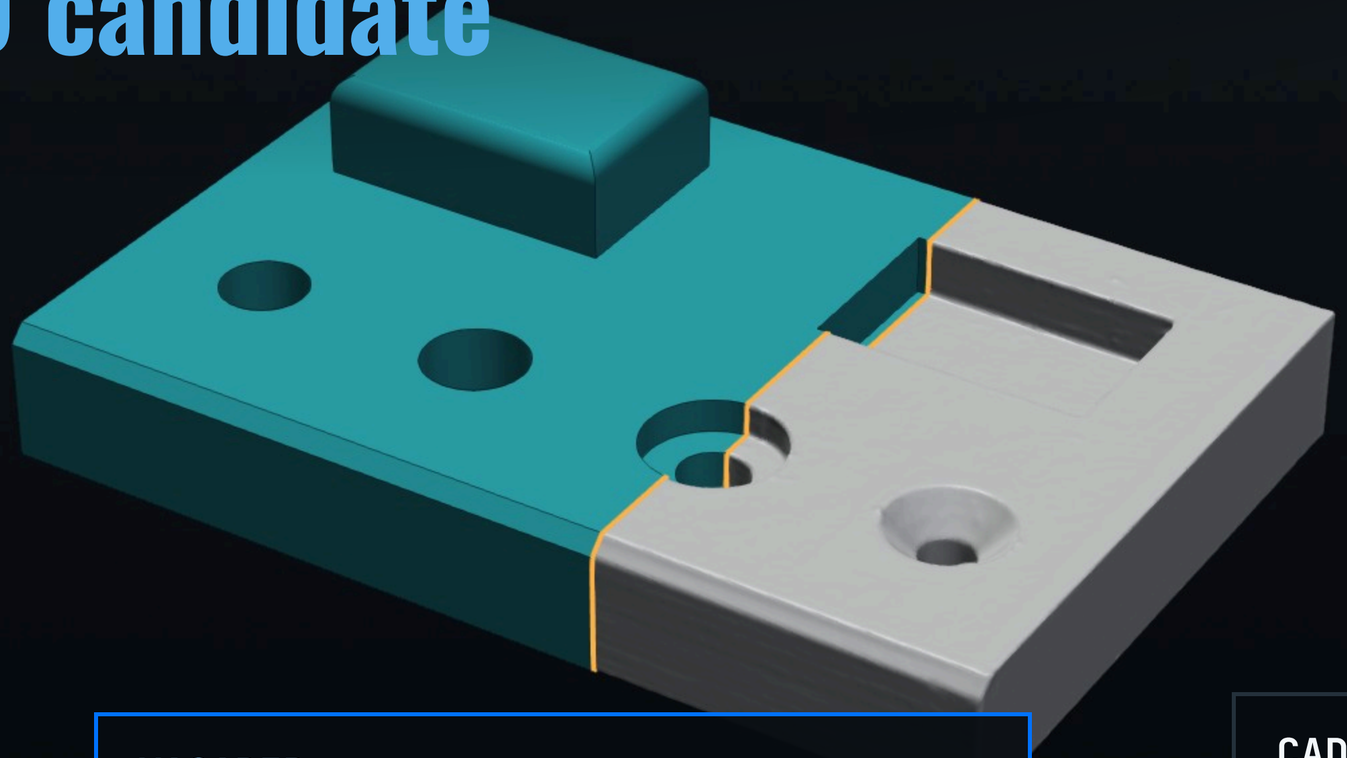
• DEVELOPMENT IN PROGRESS · WORKING PROTOTYPE · RESULTS ON A REAL SCAN

A 3D scan is not a CAD model

713,096 triangles	Dense triangles record the measured surface without a native CAD feature history.
No CAD feature tree	A hole diameter is not a native CAD parameter in the mesh.
Redrawn by hand	Engineers often rebuild the geometry as CAD features.

Our 3D-printed test part: 713,096 scan triangles, shown with mesh edges

A real scan and its native CAD candidate



REAL 3D SCAN
STL mesh

→

VACAD3D
align · repair · analyse · reconstruct · inspect

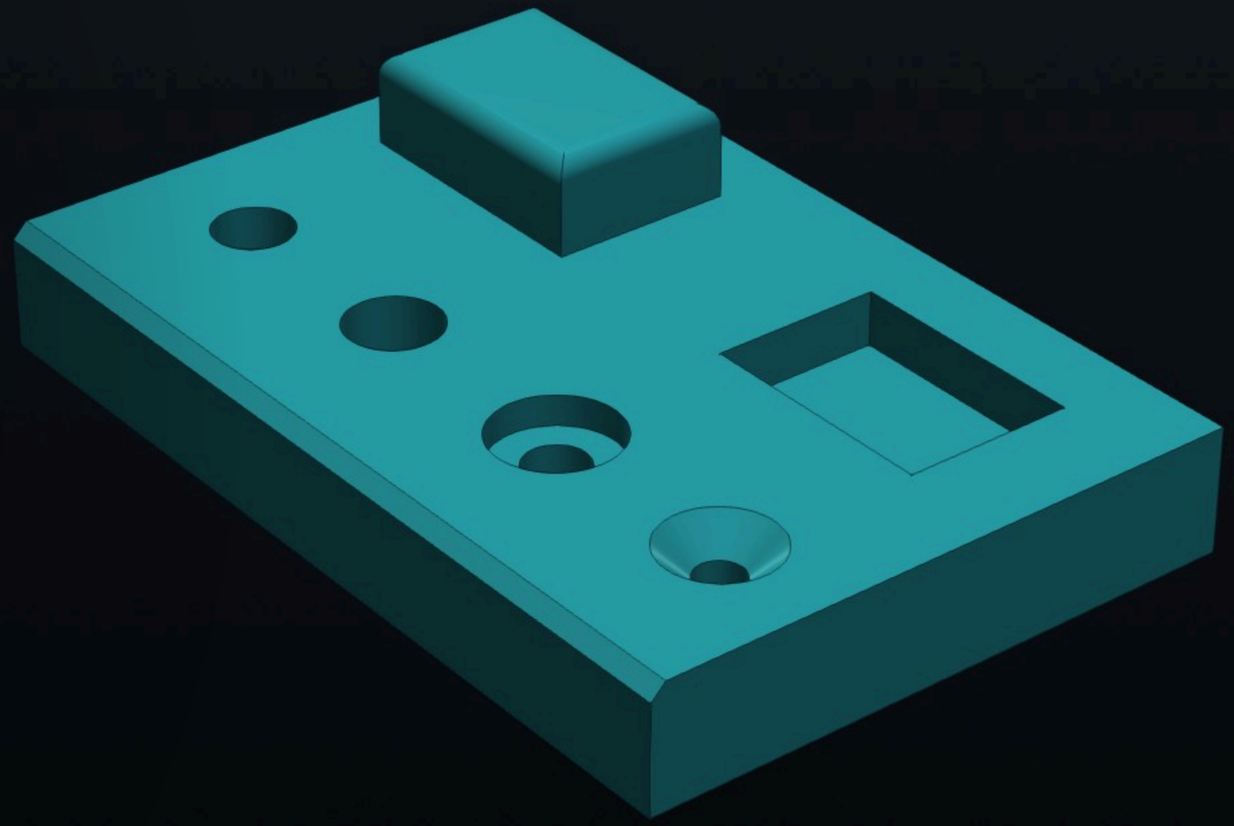
→

CAD OUTPUT
.FCStd: native feature model
STEP: solid geometry exchange

Demonstration: our own 3D-printed prismatic test part. Local fidelity review remains open.

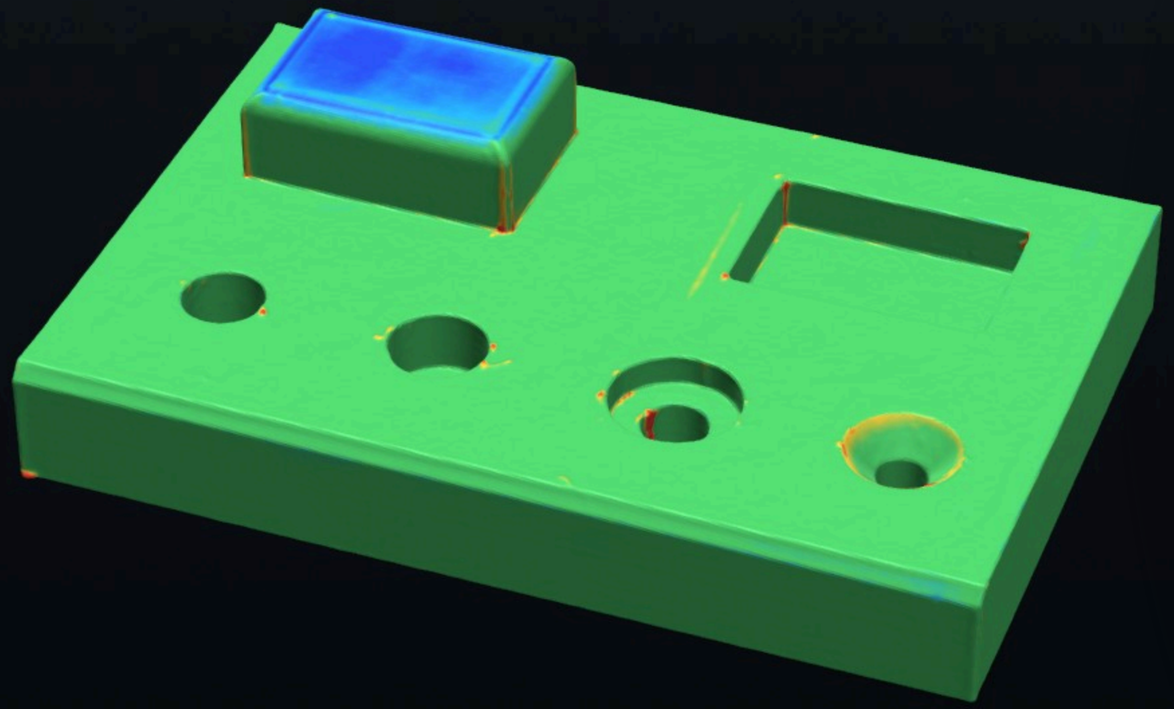
Real features.
Real dimensions

	FEATURE TREE	RECOVERED FROM SCAN
01	Pad · base	16.16 mm
02	Pad · boss	12.62 mm
03	Pocket	7.05 mm
04	Countersunk hole	Ø 6.11
05	Hole	Ø 8.10
06	Hole	Ø 10.21
07	Blind hole	Ø 12.16
08	Counterbore	Ø 16.09
09	Chamfer	1.93 mm
10	Fillet · 4 edges	R 3.18



Native FreeCAD PartDesign model: 10 features and 8 sketches

Compared with the source scan



INSPECTION · CAD vs ORIGINAL SCAN

356,544

scan vertices compared with this CAD candidate

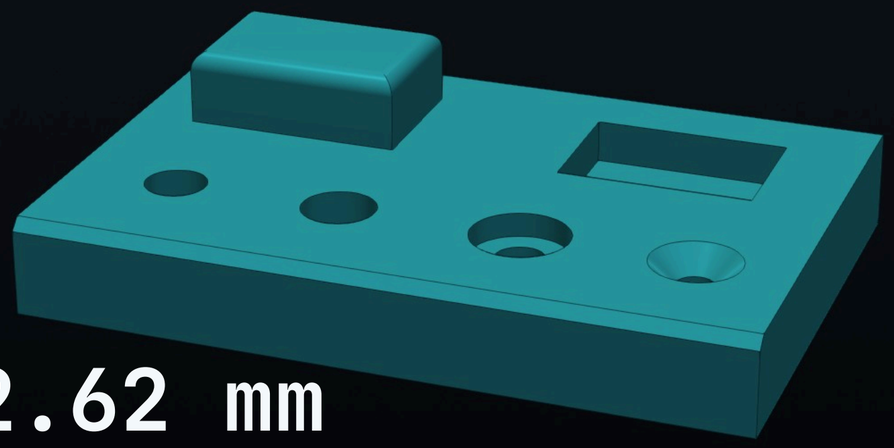
- within the display band
- model above the scan
- model below the scan

-0.3 0 mm +0.3

The colour map locates differences for engineering review. Local fidelity review remains open. Colours saturate beyond the displayed range.

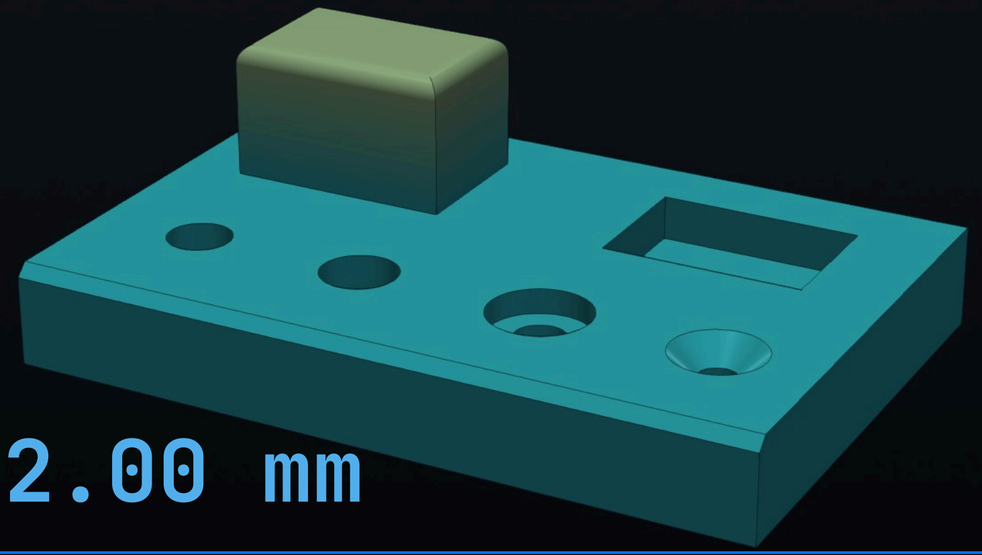
Change a number.
The model rebuilds

AS RECONSTRUCTED · BOSS HEIGHT



12.62 mm

ONE EDIT LATER · BOSS HEIGHT

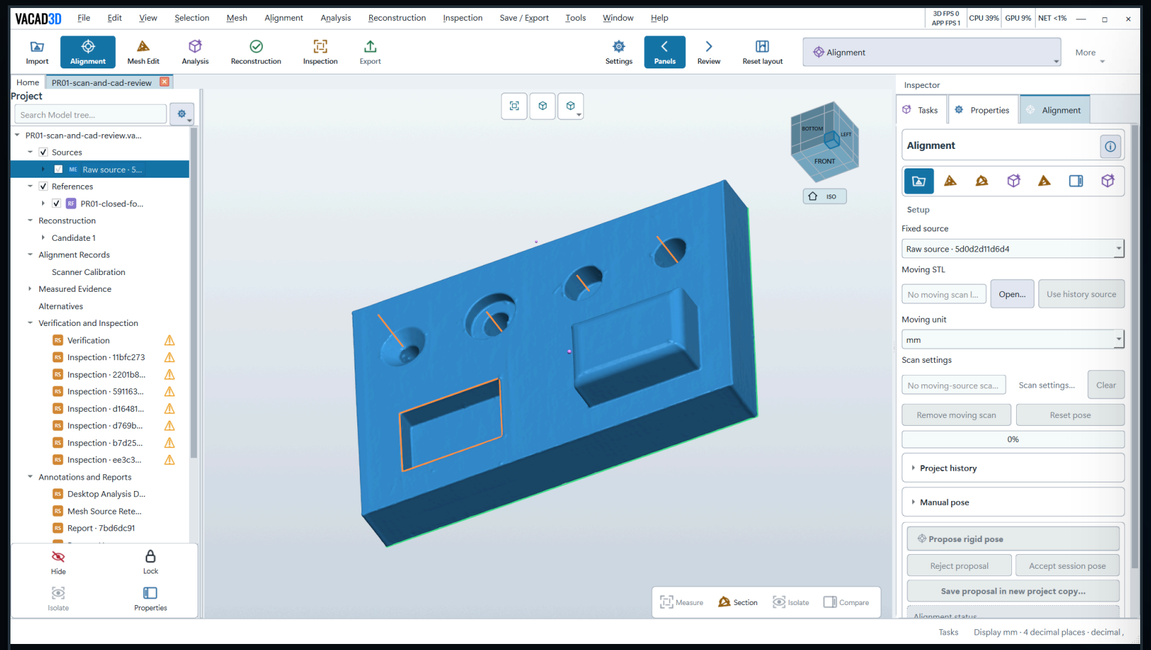


22.00 mm

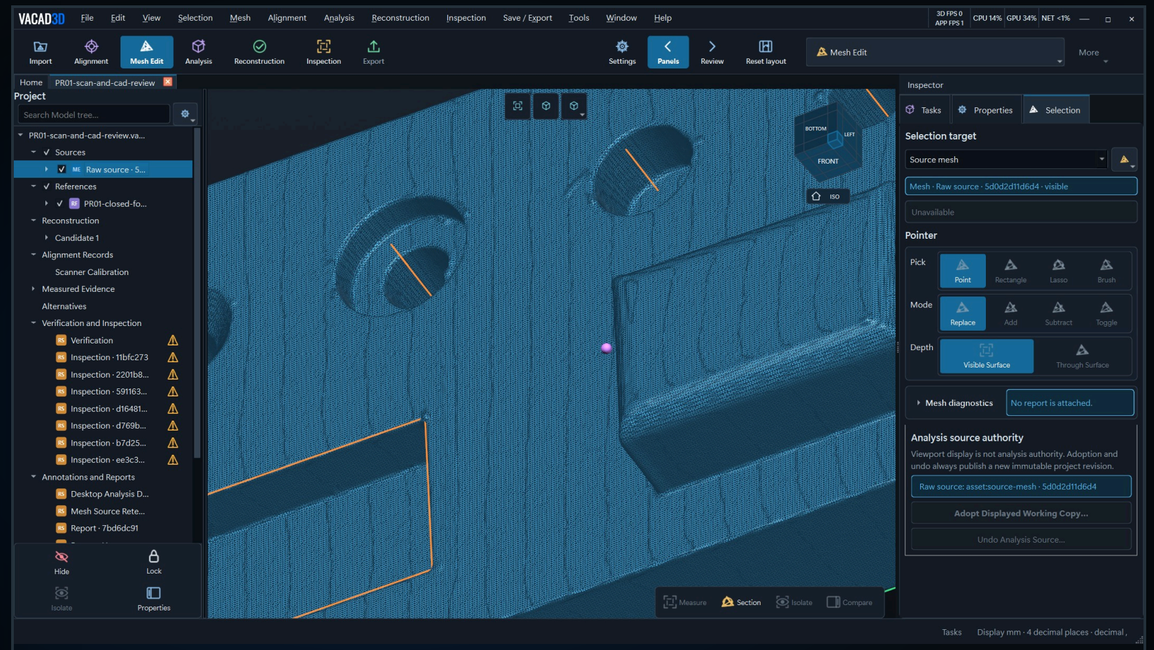
Two edits demonstrated in FreeCAD: boss height from 12.62 to 22.00 mm, and countersunk-hole diameter from 6.11 to 10.00 mm. The demonstrated states recompute as valid solids.

Edited in FreeCAD 1.1 on the exported VACAD3D model

A desktop scan-to-CAD application



LIGHT: ALIGNMENT



DARK: MESH EDITING

Real application captures with the retained scan and CAD candidate.
Light and dark themes shown in complete, uncropped workspaces.

Workspace views do not establish a completed reconstruction or inspection workflow.

Adrian, founder of VACAD3D

VACAD3D

Building a desktop application that connects measured geometry with editable engineering features.

vacad.adrian@gmail.com

Product focus

Native CAD feature history, scan comparison and FreeCAD / STEP handoff.

What the pilot must establish

Whether the workflow reduces engineering effort while meeting an agreed acceptance standard.

Who Adrian wants to meet

An industry partner with suitable parts and an investor interested in product development and qualification.

Current demonstration: our own printed test part. Customer value remains to be measured.

Engineering progress

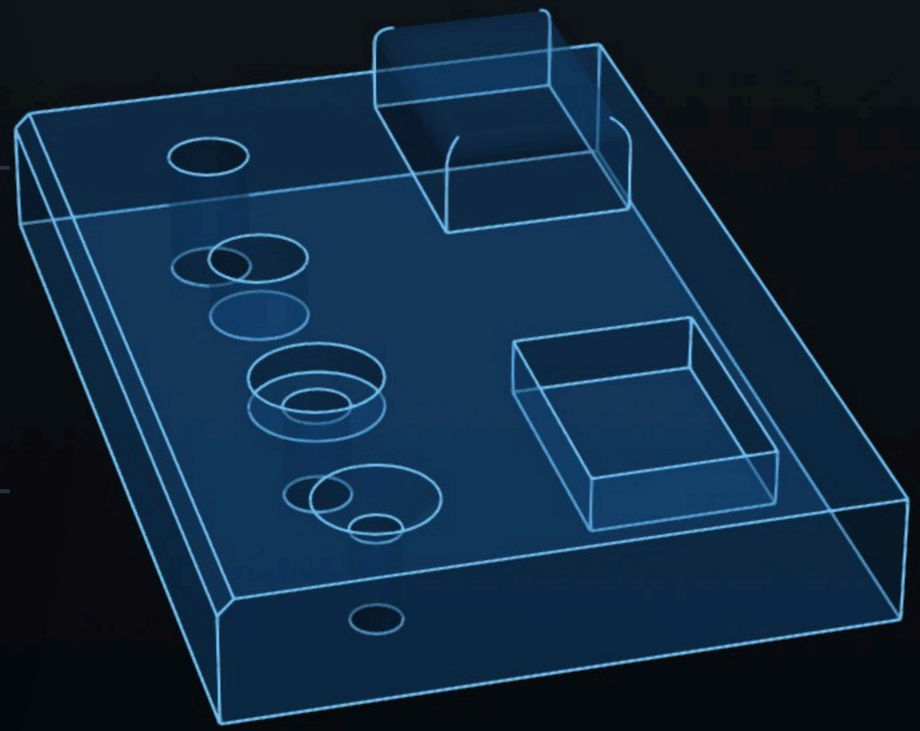
• DEVELOPMENT IN PROGRESS

9,000+
test cases in the repository

440+
commits

700,000+
lines of product code

660,000+
lines of test code



Repository size describes engineering work, not a current test pass or customer traction.

Prismatic parts without usable CAD

Proposed first partner: an engineering team rebuilding a part from a scan.

01

Select the part

Agree input requirements, data rights and the CAD deliverable before starting.

02

Agree acceptance

Review scan uncertainty, local fidelity, editable features and export requirements.

03

Measure the work

Compare operator time, including review and correction, with manual remodelling at the same acceptance scope.

Suitable candidates may include fixtures, tooling or replacement parts.
Manufacturing decisions require engineering review.

Initial pilot focus; the wider product roadmap includes broader part coverage.

Demonstrated results and next steps

Demonstrated on this test part

- ✓ A real scan reconstructed as a 10-feature FreeCAD model
- ✓ Boss-height and hole-diameter edits demonstrated in FreeCAD
- ✓ Scan and candidate CAD displayed in VACAD3D
- ✓ Local fidelity review remains open

Next

01 Resolve local fidelity concerns on the test part

02 Complete and validate the in-app edit, save and reopen workflow

03 A measured industry pilot with agreed acceptance criteria and an operator-time comparison

This candidate is not yet accepted or qualified. This demonstration covers a prismatic test part, while the product roadmap is broader. Customer time savings remain a hypothesis to test.

Discuss an investment and pilot partnership

Help develop and qualify the product, then measure its value on suitable real parts.

Development priorities

Complete application workflows, resolve fidelity concerns and qualify broader part coverage.

First conversation

Identify a suitable pilot part and agree the deliverable, acceptance criteria and comparison with manual remodelling.

VACAD3D

Adrian · Founder, VACAD3D

vacad.adrian@gmail.com

Get in touch with Adrian, the founder of VACAD3D.